AI-Driven Hiring Risk and Early Attrition Prediction Module

R26-IT-082

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Employee turnover shortly after hiring is a major challenge for organizations, leading to significant financial and operational losses. Studies show that replacing an employee can cost 150–200% of their annual salary, considering recruitment, onboarding, training, and productivity losses. Although many organizations use digital recruitment platforms and Applicant Tracking Systems (ATS), most existing systems focus only on matching candidate skills with job requirements. They rarely evaluate the risk of early employee resignation, which often results from poor role alignment, unrealistic expectations, or unstable career patterns.

This research proposes an AI-driven Early Attrition Risk Prediction module within an intelligent recruitment framework. The system analyzes candidate attributes such as employment history, frequency of job changes, qualification–role mismatch, compensation expectations, and interview insights. Machine learning models will estimate the probability of early resignation, while Explainable AI techniques such as SHAP will provide interpretable explanations for predictions.

The system aims to help recruiters identify high-risk hires early, improve hiring decisions, reduce recruitment costs, and support sustainable workforce management aligned with SDG 8: Decent Work and Economic Growth.

Keywords: Employee Attrition Prediction, Recruitment Analytics, Explainable AI, Talent Acquisition Systems, Retention Risk

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Working on this final-year research project has been a highly valuable and enriching academic experience. Throughout the development of this research, we have gained significant knowledge in identifying research problems, formulating research questions, designing appropriate methodologies, and addressing potential implementation challenges. This project has provided us with an excellent opportunity to enhance our analytical, technical, and problem-solving skills while applying theoretical knowledge in a practical research context.

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Finally, we appreciate the opportunity provided by the Sri Lanka Institute of Information Technology (SLIIT) to conduct this research as part of our undergraduate studies. We look forward to further strengthening our research capabilities and contributing meaningful knowledge through this project.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
XAI	Explainable AI
ATS	Applicant Tracking System
HR	Human Resources
ROC	Receiver Operating Characteristic
GCP	Google Cloud Platform
AUC	Area Under ROC Curve
WBS	Work Breakdown Structure
SDG	Sustainable Development Goals

Table 1:List of Abbreviations

1. INTRODUCTION

1.1 Background & Literature survey

The High employee turnover is a significant challenge for organizations because replacing an employee can cost up to one to two times the employee's annual salary due to recruitment, training, and productivity losses [1]. Many organizations experience early employee resignations caused by poor job-role alignment, unrealistic expectations, or limited career growth. Recent statistics show voluntary quit rates exceeding 2.5% per month, resulting in annual turnover rates above 15% in many sectors [8]. Early attrition disrupts productivity, increases recruitment costs, and leads to knowledge loss within organizations.

Beyond financial impacts, employee turnover also affects organizational stability by reducing employee morale, delaying projects, and weakening workforce planning. Although modern recruitment systems and Applicant Tracking Systems (ATS) help identify candidates with the right skills, they mainly focus on job–skill matching and rarely evaluate whether candidates are likely to remain in the organization long term. Most existing attrition prediction models rely on post-hire data such as employee satisfaction surveys, salary information, and performance records [2], [7], making them ineffective during the hiring stage.

Additionally, many machine learning models used for attrition prediction operate as black-box systems, providing predictions without explaining the factors behind them [4], [5]. This lack of transparency limits trust and usability in HR decision-making. To address these limitations, this research proposes an AI-driven retention intelligence system that predicts early resignation risk using candidate data available during recruitment, including resumes, employment history, and interview insights.

The proposed system will apply machine learning algorithms such as Random Forest, XGBoost, or neural networks to predict attrition risk. It will also integrate Explainable AI techniques such as SHAP and LIME to identify key factors influencing predictions [10], [11]. By providing transparent risk scores and explanations, the system enables recruiters to make more informed hiring decisions and proactively reduce early employee turnover.

Furthermore, the project contributes to Sustainable Development Goal 8 (Decent Work and Economic Growth) by supporting sustainable workforce management and aligns with the Data Science and AI research cluster (IT4010) by advancing explainable machine learning applications in HR analytics.

Literature Survey

Employee attrition has been widely studied in the field of Human Resource Analytics and Machine Learning, as organizations increasingly rely on data-driven methods to understand workforce dynamics and predict employee turnover. Recent research has focused on developing predictive models that identify factors influencing employee resignations and support proactive workforce management strategies [2], [7].

Several studies have applied machine learning algorithms to predict employee attrition using historical HR datasets. For example, Raza et al. (2022) conducted a comparative study of multiple machine learning algorithms, including Decision Trees, Random Forest, Support Vector Machines, and Extra Trees classifiers, to predict employee attrition using the IBM HR Analytics dataset. Their results demonstrated that ensemble-based algorithms achieved higher predictive accuracy, reaching approximately 93% accuracy in predicting employee turnover. The study also identified key factors influencing attrition such as work environment, salary levels, and career development opportunities [2].

Similarly, Krishna and Sidharth (2022) developed a Random Forest-based attrition prediction model using HR analytics data. Their research highlighted that machine learning can effectively identify employees at risk of leaving by analyzing attributes such as job role, job satisfaction, workload, and promotion history. The study demonstrated that predictive analytics can help organizations take proactive measures to improve employee retention and reduce recruitment costs [2], [6].

Another line of research focuses on comparing different machine learning models to determine the most effective techniques for attrition prediction. Govindarajan (2025) evaluated several algorithms including Random Forest, Support Vector Machine, Decision Tree, Gradient Boosting, and hybrid models. The results showed that ensemble methods often outperform individual classifiers due to their ability to capture complex relationships between employee characteristics and attrition

outcomes [6].

Recent research has also explored the use of Explainable Artificial Intelligence (XAI) to improve transparency in attrition prediction models. Baydili and Tasci (2025) proposed an XAI-powered predictive framework that combines machine learning models with explainability techniques to identify key factors influencing employee turnover. Their approach demonstrated that explainable models provide better interpretability for HR professionals and improve trust in AI-based decision-support systems [4].

In addition to traditional machine learning approaches, emerging studies investigate the use of advanced artificial intelligence techniques such as deep learning and large language models for predicting employee attrition. Studies comparing advanced AI models with traditional machine learning classifiers have shown that fine-tuned models can achieve improved predictive performance with higher precision and recall scores, suggesting that advanced AI techniques can identify complex patterns in workforce behavior and organizational data [7], [8].

Beyond academic research, several commercial HR analytics platforms have incorporated predictive analytics capabilities to assist organizations in workforce planning and retention management. For instance, the **SAP SuccessFactors People Analytics** platform provides HR analytics tools that allow organizations to analyze workforce trends, track employee engagement, and identify potential attrition risks using predictive analytics techniques [12].

Similarly, **Workday People Analytics** integrates machine learning and workforce data to generate insights into employee engagement and retention. By combining employee sentiment data with human capital management metrics, the system helps HR leaders understand factors influencing workforce stability and make data-driven decisions regarding talent management [13].

Despite these advancements, several limitations remain in existing research and systems. First, most attrition prediction models rely heavily on post-hire employee data, such as job satisfaction surveys, salary information, and performance evaluations. While these models can identify employees who are likely to leave, they do not assist recruiters during the pre-hiring stage, when critical hiring decisions are made [2], [8].

Second, many machine learning models used in HR analytics operate as black-box systems, making

it difficult for HR professionals to interpret predictions and understand the underlying factors influencing attrition risk. Explainability is therefore becoming an important requirement for AI systems used in human resource decision-making [4], [10], [11].

Another limitation is that many commercial HR analytics tools focus primarily on workforce trend analysis rather than individual-level prediction for new hires. Although platforms such as SAP SuccessFactors and Workday provide valuable insights into employee retention and workforce performance, their predictive capabilities are typically applied to existing employees rather than evaluating candidates before hiring [12], [13].

Furthermore, many academic research prototypes lack user-friendly interfaces and integration with recruitment systems, limiting their adoption in real-world HR workflows. Recruiters require practical decision-support systems that provide not only predictions but also clear explanations and actionable recommendations [7].

Therefore, a significant research gap exists in developing **pre-hire attrition prediction systems** that combine machine learning, explainable AI, and recruiter-oriented decision-support interfaces. Addressing this gap could enable organizations to evaluate candidate retention risk before hiring and implement proactive strategies to reduce early employee turnover.

1.2 Research Gap

Although significant progress has been made in employee attrition prediction using machine learning and data analytics, several critical limitations remain in existing research and industry systems. Most current studies focus primarily on post-hire attrition prediction, where models are trained using organizational data such as employee satisfaction surveys, salary levels, performance evaluations, promotion history, and engagement metrics [2], [8]. While these models can accurately identify employees who are likely to leave an organization, they only provide insights after employees have already joined the company. As a result, these approaches support reactive retention strategies rather than enabling organizations to prevent costly turnover during the recruitment stage.

Another limitation of existing approaches is the lack of focus on pre-hire data sources. Candidate-

level information such as employment history, job-tenure patterns, qualification mismatches, and interview responses can provide valuable signals regarding long-term retention potential. However, most current predictive models do not utilize these indicators because they rely heavily on internal organizational datasets that become available only after hiring [7].

Furthermore, many machine learning models developed for attrition prediction operate as black-box systems, producing predictions without explaining the reasoning behind them. Although these models may achieve high predictive accuracy, the absence of interpretability limits their practical adoption in HR decision-making contexts. Explainable AI methods such as SHAP and LIME have been proposed to address this issue by providing interpretable explanations for model predictions [10], [11].

Existing commercial HR analytics platforms also exhibit similar limitations. Enterprise solutions such as SAP SuccessFactors and Workday People Analytics provide tools for analyzing workforce trends and identifying patterns of employee engagement or turnover. However, these systems primarily focus on analyzing existing employees rather than predicting the attrition risk of potential hires during recruitment [12], [13].

Therefore, a clear research gap exists in the development of **pre-hire attrition prediction systems that combine predictive analytics with explainable AI techniques**. A system capable of analyzing candidate-level information before hiring and generating interpretable attrition risk scores would enable organizations to make more informed recruitment decisions.

1.3 Research Problem

Employee attrition, particularly early resignation after hiring, has become a significant challenge for modern organizations. High employee turnover results in substantial financial losses due to recruitment expenses, onboarding processes, training investments, and productivity disruptions. Studies indicate that replacing an employee can cost up to one to two times the employee's annual salary [1].

Existing Human Resource analytics systems and research models focus mainly on post-hire attrition prediction using organizational data such as employee satisfaction surveys, salary records,

performance evaluations, and engagement metrics [2], [8]. Although these models can identify employees who are likely to leave after joining the organization, they do not assist recruiters during the hiring stage when critical decisions are made.

Furthermore, many machine learning models used in attrition prediction operate as black-box systems, providing predictions without clear explanations. In Human Resource decision-making, transparency is essential because recruiters must understand why a candidate is considered high-risk before making hiring decisions [4], [10].

Therefore, there is a critical need to develop an AI-driven predictive system capable of estimating employee resignation risk before hiring decisions are finalized. Such a system should analyze candidate-level data, generate accurate attrition risk predictions, and provide explainable insights that help recruiters understand the factors influencing the prediction.

Research Questions:

1. How can pre-hire candidate data be utilized to predict the risk of early employee resignation?

    - This question addresses the core research challenge of identifying meaningful features from candidate profiles, such as employment history, job tenure patterns, qualification mismatches, and interview responses.

    - It explores how these indicators can be transformed into predictive features for machine learning models capable of estimating attrition risk before hiring.

2. How can machine learning techniques be used to accurately predict employee attrition risk during the recruitment stage?

    - This question focuses on developing predictive models using algorithms such as Random Forest, Gradient Boosting, or neural networks.

    - It investigates how these models can analyze candidate data and generate reliable resignation risk predictions.

3. How can Explainable Artificial Intelligence (XAI) techniques improve transparency and trust in attrition prediction systems?
    - This question examines the use of interpretability methods such as SHAP and LIME to explain the factors influencing predictions.
    - It aims to ensure that recruiters understand why a candidate is considered high-risk, thereby improving the practical adoption of AI-based HR analytics tools.

4. How can predictive attrition systems contribute to reducing early employee turnover and improving workforce sustainability?
    - This question explores the broader organizational impact of predictive recruitment intelligence.
    - It examines how proactive attrition prediction can support better hiring strategies, reduce recruitment costs, and improve long-term workforce stability.

This research aims to address critical gaps in the development of predictive analytics systems for employee retention within the recruitment process. By exploring the above research questions, the study seeks to develop an AI-driven framework capable of analyzing pre-hire candidate data to predict the risk of early employee resignation. The research will integrate machine learning techniques with Explainable Artificial Intelligence (XAI) methods to provide accurate predictions along with transparent explanations that support recruiter decision-making. The outcomes of this research will contribute to the development of intelligent recruitment support systems that enable organizations to evaluate both candidate-job fit and long-term retention potential before hiring. Ultimately, the proposed approach aims to help organizations reduce early employee turnover, improve recruitment efficiency, and support sustainable workforce management through data-driven hiring strategies.

2. OBJECTIVES

2.1 Main Objective

The primary objective of this research is to develop an AI-driven predictive attrition risk model that can estimate the likelihood of early employee resignation during the recruitment stage. The proposed system will analyze pre-hire candidate data such as employment history, job tenure patterns, qualification mismatches, and interview responses to generate accurate resignation risk predictions. In addition, the model will integrate Explainable Artificial Intelligence (XAI) techniques to provide transparent insights into the factors influencing each prediction. The system will be designed as an intelligent decision-support tool that assists recruiters in evaluating both candidate suitability and long-term retention potential. Ultimately, the goal is to create a scalable and practical solution that helps organizations make data-driven hiring decisions, reduce early employee turnover, and improve overall workforce sustainability.

2.2 Sub Objectives

Pre-Hire Data Feature Extraction and Engineering

- Identify and analyze relevant pre-hire candidate attributes such as employment history, job tenure patterns, educational background, and qualification-to-job mismatch indicators that may influence early employee resignation.Fine-tune the chosen model on a curated dataset that includes mathematical problems, solution steps, and explanatory content to ensure the chatbot can provide accurate and pedagogically sound guidance.
- Develop a feature engineering framework to transform raw candidate information from resumes, application forms, and interview notes into structured variables suitable for machine learning analysis.
- Evaluate the significance of different features using statistical and machine learning techniques to determine which candidate characteristics contribute most strongly to predicting early attrition risk.

Development and Optimization of the Predictive Attrition Model

- Implement and compare multiple machine learning algorithms, such as Random Forest, Gradient Boosting (XGBoost), and Neural Networks, to identify the most effective approach for predicting employee resignation risk.Integrate features such as real-time feedback, hints, and visual aids (e.g., graphs, charts) to enhance the user's understanding of complex mathematical concepts.
- Train and fine-tune the selected model using historical HR datasets to improve predictive accuracy and ensure reliable estimation of attrition probabilities.
- Evaluate model performance using metrics such as Accuracy, F1-Score, and ROC-AUC to ensure the model effectively identifies candidates with higher risk of early resignation.

Integration of Explainable Artificial Intelligence (XAI)

- Apply explainability techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) to interpret machine learning predictions and identify key factors contributing to attrition risk.Optimize the computational efficiency of the language model to minimize operational costs, making the solution affordable for widespread use, particularly in resource- constrained environments.
- Develop mechanisms to generate clear explanations that highlight influential features, such as frequent job changes, overqualification, or mismatch between candidate skills and job requirements.
- Assess the effectiveness of explainability methods in improving transparency and trust among HR professionals when using AI-driven decision-support tools.

Design and Development of a Recruitment Decision-Support System

- Design a user-friendly web-based dashboard that allows recruiters to upload candidate data, view attrition risk predictions, and access explanation insights generated by the model.
- Integrate the predictive model with a backend system capable of processing candidate data

and generating risk scores in real time.

- Implement visualization tools that present predictions and explanatory insights in a clear and actionable format to support recruiter decision-making.

Evaluation of System Effectiveness and Organizational Impact

- Conduct experimental testing to evaluate the predictive accuracy, efficiency, and reliability of the proposed attrition prediction system.
- Perform usability testing with potential users (e.g., HR professionals or recruiters) to assess the practicality and usefulness of the system in real recruitment scenarios.
- Analyze how predictive attrition insights can support proactive hiring strategies and contribute to reducing early employee turnover and improving workforce sustainability.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach combined with experimental evaluation. The Design Science methodology is appropriate because the study aims to design and implement a technological artifact—an AI-driven predictive system that estimates employee resignation risk during the recruitment stage. The research process involves identifying the problem, designing the predictive model, implementing the system prototype, and evaluating its effectiveness using empirical data.

The study will also include experimental analysis to compare different machine learning algorithms and determine the most accurate model for predicting early employee attrition. The system will be iteratively improved based on evaluation results and usability feedback to ensure its effectiveness as a recruitment decision-support tool.

3.2 Technical Methods and Algorithms

The technical methodology involves several stages including data preprocessing, feature engineering, predictive modeling, and explainability integration.

First, candidate and employee datasets will be preprocessed to remove inconsistencies, handle missing values, and convert categorical attributes into machine-readable formats using encoding techniques such as One-Hot Encoding and Label Encoding.

Feature engineering will be applied to extract meaningful predictors related to employee attrition risk. These features may include employment tenure patterns, number of previous job changes, qualification-to-job mismatch indicators, and candidate experience levels.

Multiple machine learning algorithms will be implemented and compared, including:

- Random Forest Classifier – for robust ensemble learning and handling complex feature interactions.
- Gradient Boosting / XGBoost – for improved predictive accuracy and handling non-linear relationships in HR datasets.
- Artificial Neural Networks (ANN) – for modeling complex patterns and relationships in candidate data.

To improve transparency and interpretability, the system will incorporate Explainable Artificial Intelligence (XAI) techniques such as:

- SHAP (SHapley Additive Explanations) – to quantify the contribution of each feature to a prediction.
- LIME (Local Interpretable Model-Agnostic Explanations) – to provide local explanations for individual candidate predictions.

These methods will allow recruiters to understand why a candidate is predicted to have a higher attrition risk.

3.3 Tools and Platforms

The system will be developed using modern data science and web development tools that support machine learning experimentation and system deployment.

- Python Programming Language – Python is widely used in data science and machine learning due to its extensive libraries and community support.
- Scikit-learn – used for implementing machine learning algorithms such as Random Forest and classification models.
- XGBoost Library – used for gradient boosting models with high predictive performance.
- SHAP and LIME Libraries – used for explainable AI techniques to interpret model predictions.
- Pandas and NumPy – used for data preprocessing, transformation, and statistical analysis.
- Streamlit or React + Flask – used to develop a user-friendly dashboard for recruiters to interact with the prediction system.
- PostgreSQL or MongoDB – used as the backend database to store candidate data and prediction results.

These tools are selected because they are open-source, scalable, and widely used in industry and research, making them suitable for developing an AI-based HR analytics system.

3.4 Data Collection and Ethical Considerations

The research will primarily use publicly available datasets for employee attrition analysis, such as the IBM HR Analytics Employee Attrition dataset available on Kaggle. This dataset contains employee-related attributes including job role, salary, tenure, job satisfaction, and attrition status, which can be used to train predictive models.

Additional synthetic or simulated candidate data may be generated to represent pre-hire recruitment attributes, such as resume information and employment history.

Ethical considerations will be carefully addressed throughout the research process. Personal identifiers will be removed to ensure data anonymization and privacy protection. Sensitive attributes such as gender, race, or ethnicity will not be used as predictive features to prevent algorithmic bias

or discrimination.

Furthermore, the system will be designed as a decision-support tool rather than an automated decision-making system, ensuring that final hiring decisions remain under human control.

3.5 Validation Strategy and Performance Metrics

To evaluate the effectiveness of the proposed predictive model, the dataset will be divided into training and testing subsets using techniques such as k-fold cross-validation to ensure robust model evaluation.

The performance of the machine learning models will be measured using several classification metrics, including:

- Accuracy – measures the overall correctness of predictions.
- Precision – measures how accurately the model identifies true attrition cases.
- Recall (Sensitivity) – measures the ability of the model to detect employees who are likely to resign.
- F1-Score – provides a balanced evaluation of precision and recall.
- ROC-AUC (Receiver Operating Characteristic – Area Under Curve) – measures the model's ability to distinguish between attrition and non-attrition cases.

In addition to predictive performance, the effectiveness of explainability techniques will be evaluated by analyzing how clearly the system presents feature contributions and whether these insights improve user understanding of model predictions.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed system is an AI-driven predictive attrition risk analysis platform designed to help recruiters identify candidates who may have a high probability of early resignation. The system analyzes pre-hire candidate data such as resumes, employment history, qualifications, and interview responses to detect patterns that may indicate potential attrition risks. Machine learning algorithms, including Random Forest, XGBoost, and Neural Networks, are used to estimate the likelihood of early employee turnover.

To improve transparency and trust, the platform integrates Explainable Artificial Intelligence (XAI) techniques such as SHAP to interpret model predictions and highlight the factors contributing to the attrition risk score. The system follows a modular architecture consisting of components for data ingestion, preprocessing and feature engineering, predictive modeling, explainability analysis, data storage, and a recruiter dashboard interface.

These components interact through a structured data pipeline that enables efficient processing, prediction, and visualization of candidate attrition risk. The final results are presented through a recruitment dashboard, allowing recruiters to make data-driven hiring decisions and proactively reduce early employee turnover.

4.2 High-Level System Architecture Diagram

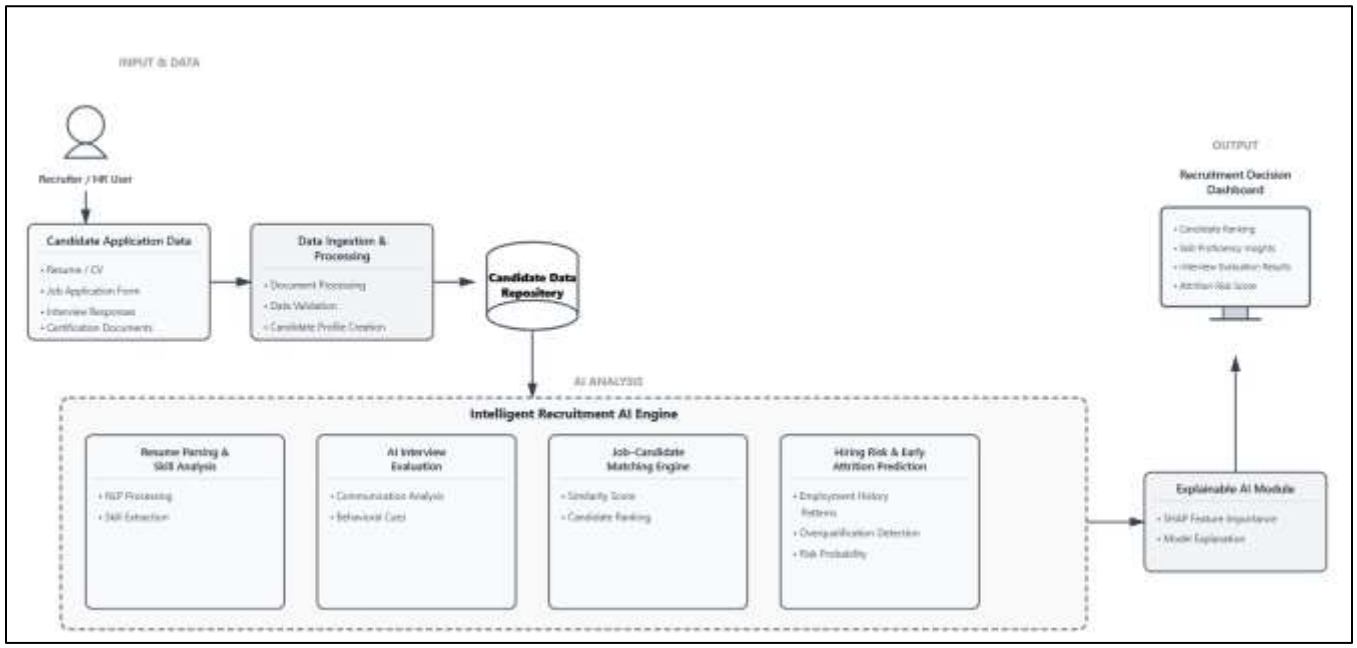


Figure 1 : AI-Driven Intelligent Talent Acquisition System Architecture

Figure 1 illustrates the high-level architecture of the proposed AI-Driven Intelligent Recruitment Decision Platform. The system is organized into three main layers: Input & Data Layer, AI Analysis Layer, and Output & Decision Layer. The platform collects candidate information, processes the data, applies artificial intelligence techniques for analysis, and delivers intelligent hiring insights to recruiters.

The process begins with the Recruiter / HR User, who uploads candidate information such as resumes, job application forms, interview responses, and certification documents through the system interface. This data is processed by the Data Ingestion & Processing module, which performs document processing, validation, and conversion of raw inputs into structured candidate profiles. The processed data is then stored in the Candidate Data Repository, which acts as the centralized storage system for recruitment data.

The Intelligent Recruitment AI Engine serves as the core analytical component of the platform. It includes several AI modules such as Resume Parsing & Skill Analysis, AI Interview Evaluation, Job-Candidate Matching Engine, and Hiring Risk & Early Attrition Prediction. These modules analyze candidate qualifications, evaluate interview responses, calculate job-candidate similarity scores, and estimate the risk of early employee resignation.

To ensure transparency, the system integrates an Explainable AI (XAI) module, which uses techniques such as SHAP to identify and explain the factors influencing AI predictions. Finally, the analyzed results are presented through the Recruitment Decision Dashboard, where recruiters can view candidate rankings, skill insights, interview evaluations, and attrition risk scores.

Overall, the system enables recruiters to make data-driven hiring decisions by transforming candidate information into meaningful insights through an integrated AI-powered recruitment workflow.

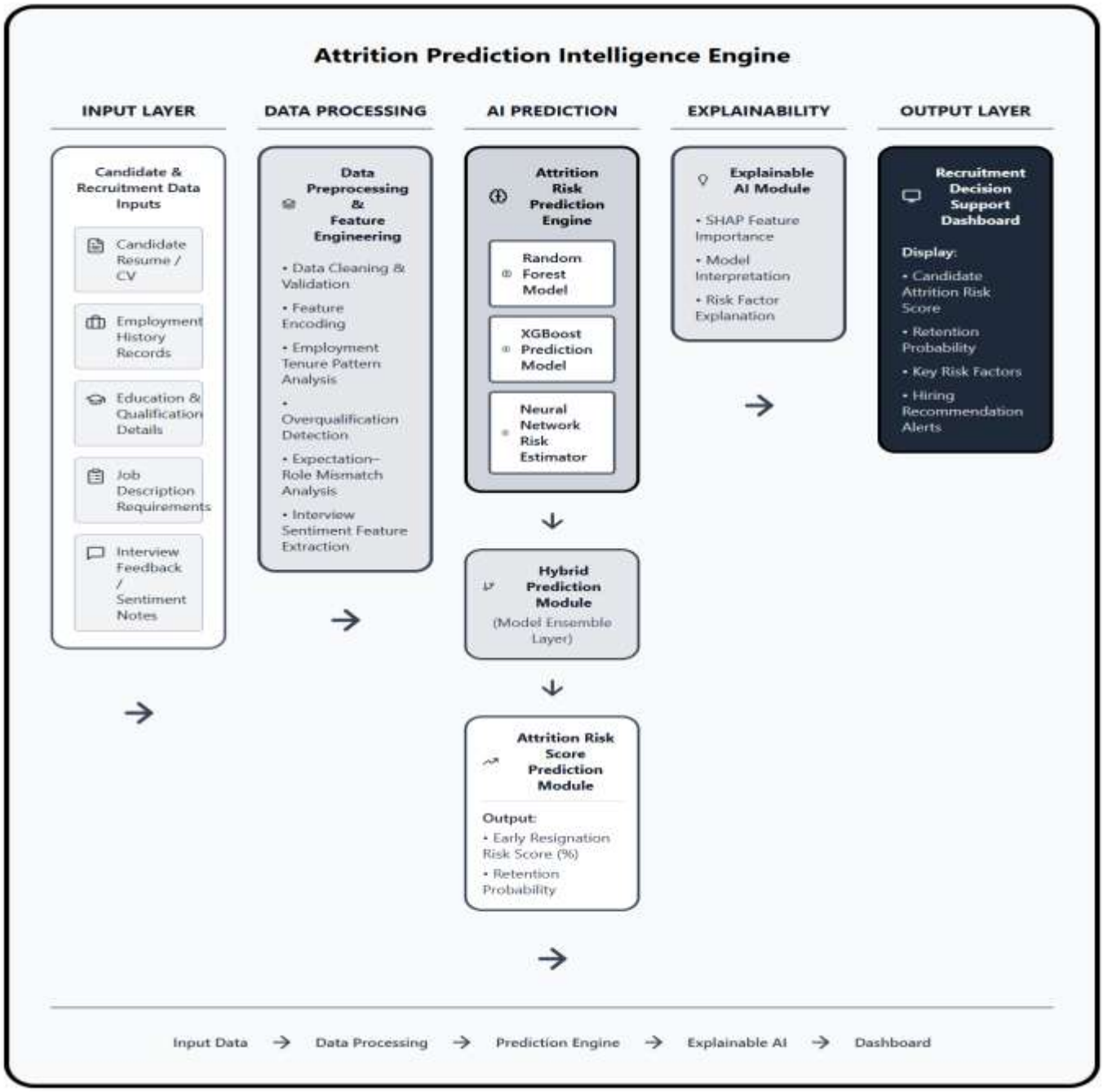


Figure 2 : System Architecture for Attrition Prediction Model

Figure 2 illustrates the architecture of the Hiring Risk & Early Attrition Prediction Model, which is responsible for predicting the likelihood of early employee resignation during the recruitment process. The architecture consists of five main layers: Input Layer, Data Processing Layer, AI Prediction Layer, Explainability Layer, and Output Layer.

The system begins with the Candidate & Recruitment Data Inputs, which include resumes, employment history records, educational qualifications, job description requirements, and interview feedback. This information is processed in the Data Preprocessing & Feature Engineering module, where data cleaning, validation, and feature extraction are performed to generate structured inputs for machine learning models.

The processed data is then analyzed by the Attrition Risk Prediction Engine, which uses machine learning algorithms such as Random Forest, XGBoost, and Neural Networks. These models work together within a hybrid prediction framework to generate an Attrition Risk Score and estimate the candidate's retention probability.

To improve transparency, the system incorporates an Explainable AI module, which uses techniques such as SHAP to identify key factors influencing the predicted attrition risk. Finally, the results are presented through the Recruitment Decision Support Dashboard, where recruiters can view attrition risk scores, retention probabilities, and key risk indicators to support data-driven hiring decisions.

4.3 Key System Components

1. Recruiter Dashboard (User Interface)

This component provides a web-based interface that allows recruiters to interact with the system. Recruiters can upload candidate profiles, view predicted attrition risk scores, and analyze explanations generated by the AI system. The dashboard presents results through visualizations such as charts and feature importance graphs to support informed hiring decisions.

2. Data Ingestion Module

The data ingestion component collects candidate information from various sources such as resumes, application forms, and interview transcripts. This module converts uploaded documents into structured datasets that can be processed by the machine learning system.

Key functions include:

- Resume parsing
- Candidate data extraction
- API-based data submission

3. Data Preprocessing and Feature Engineering

This component prepares raw candidate data for machine learning analysis. It performs tasks such as cleaning missing values, converting categorical variables into numerical representations, and generating meaningful predictive features.

Important engineered features include:

- Number of previous job changes
- Average employment tenure
- Qualification-job mismatch indicators
- Experience level metrics

4. Machine Learning Prediction Engine

The prediction engine applies trained machine learning models to estimate the probability that a candidate may resign early after joining the organization.

The system will evaluate multiple algorithms including:

- Random Forest
- Gradient Boosting (XGBoost)
- Artificial Neural Networks

The best-performing model will generate the final attrition risk probability score.

5. Explainable AI (XAI) Module

To improve transparency and trust, the system integrates explainable AI techniques that interpret machine learning predictions.

Methods used include:

- SHAP (SHapley Additive Explanations) to determine the contribution of each feature.
- LIME (Local Interpretable Model-Agnostic Explanations) to explain individual predictions.

These explanations help recruiters understand the reasons behind predicted attrition risks.

6. Database and Storage Layer

The database component stores candidate data, prediction results, feature values, and system logs. It enables efficient retrieval of historical predictions and supports continuous model improvement.

Technologies such as PostgreSQL or MongoDB can be used to manage structured and semi-structured HR data.

4.4 Component Interactions

The proposed system consists of several interconnected components that collaborate to analyze candidate data and generate attrition risk predictions. Each component performs a specific function within the overall workflow while exchanging data with other modules to ensure seamless system operation.

The interaction process begins with the Recruiter / HR User, who uploads candidate information such as resumes, employment records, and interview feedback through the Recruitment Dashboard Interface. This information is transmitted to the Data Ingestion Module, which collects and organizes the raw candidate data from different input sources.

The ingested data is then forwarded to the Data Preprocessing and Feature Engineering Module, where the raw data is cleaned, validated, and transformed into structured features suitable for

machine learning analysis. During this stage, important attributes such as employment tenure patterns, qualification mismatches, and interview sentiment indicators are extracted.

Once the data is processed, the structured feature set is passed to the Attrition Risk Prediction Engine, which contains multiple machine learning models including Random Forest, XGBoost, and Neural Networks. These models analyze the candidate features and generate a predicted probability representing the candidate's likelihood of early resignation.

The prediction results are then sent to the Explainable AI (XAI) Module, which analyzes the model output and identifies the most influential features contributing to the prediction. Techniques such as SHAP are used to provide transparent explanations of the model's decisions.

Finally, the generated predictions and explanations are delivered to the Recruitment Decision Dashboard, where recruiters can view the attrition risk score, retention probability, and key risk factors for each candidate. All candidate data, prediction outputs, and system logs are also stored in the Database and Storage Module to support data persistence and future analysis.

Through this interaction workflow, the system ensures that candidate data flows efficiently between components, enabling accurate attrition predictions and providing interpretable insights that support informed recruitment decisions.

4.5 Data Flow Explanation

The data flow begins when recruiters upload candidate information such as resumes, employment history, qualifications, and interview feedback through the recruitment system interface. The uploaded data is processed by the data ingestion module, which collects and organizes candidate information from different sources. This data is then passed to the data preprocessing and feature engineering module, where it is cleaned, validated, and transformed into structured features such as employment tenure patterns, qualification–job mismatch indicators, and interview sentiment scores.

The processed features are then provided to the Attrition Risk Prediction Engine, which applies machine learning algorithms such as Random Forest, XGBoost, and Neural Networks to estimate the

probability of early employee resignation. The predicted results are analyzed by the Explainable AI (XAI) module, which identifies the key factors influencing the prediction using techniques such as SHAP. Finally, the results, including the attrition risk score, retention probability, and key risk factors, are stored in the database and displayed to recruiters through the Recruitment Decision Dashboard, enabling informed hiring decisions.

4.6 Scalability and Security Considerations

To ensure that the system can operate effectively in real-world recruitment environments, both scalability and security measures are incorporated into the architecture.

Scalability considerations include:

- Cloud-based deployment to support large-scale recruitment operations and data processing.
- Modular system architecture that allows independent scaling of components such as the prediction engine and data processing modules.
- Efficient machine learning inference pipelines capable of analyzing large volumes of candidate data in real time.

Security considerations include:

- Encryption of candidate data during transmission and storage to protect sensitive personal information.
- Role-based access control to ensure that only authorized HR personnel can access recruitment data and system insights.
- Anonymization of sensitive attributes to prevent bias and protect candidate privacy.
- Compliance with ethical AI practices to ensure fairness and transparency in AI-driven hiring decisions.

These considerations ensure that the system remains secure, scalable, and reliable, while enabling organizations to apply AI-driven recruitment intelligence to large candidate datasets efficiently.

5. PROJECT REQUIREMENTS

The development of the proposed AI-Driven Intelligent Recruitment Decision Platform requires careful consideration of both functional and non-functional requirements. These requirements ensure that the system effectively supports recruitment decision-making, operates efficiently, and maintains reliability, security, and scalability.

Functional requirements define the core capabilities of the system, while non-functional requirements ensure that the system performs reliably under real-world recruitment conditions. In addition, system and user requirements are defined to guarantee that the platform can be successfully developed, deployed, and used by HR professionals.

5.1 Functional requirements

1. **Candidate Data Processing**
   - The system should allow recruiters to upload candidate information including resumes, employment history records, interview feedback, and qualification details.
   - The system must automatically process and organize candidate data to create structured candidate profiles suitable for AI analysis.

2. **Resume Parsing and Skill Extraction**
   - The system should analyze candidate resumes using Natural Language Processing (NLP) techniques to extract key information such as skills, experience, educational qualifications, and certifications.
   - Extracted information should be used to identify candidate competencies relevant to job requirements.

3. **Job–Candidate Matching**

- The system must compare candidate profiles with job description requirements to calculate a similarity score.
- Based on this score, the system should rank candidates according to their suitability for the job role.

4. **Early Attrition Risk Prediction**

- The system should predict the likelihood of early employee resignation using machine learning models such as Random Forest, XGBoost, or Neural Networks.
- The prediction should generate an attrition risk probability based on candidate employment history patterns, overqualification indicators, and job expectation mismatches.

5. **Explainable AI Insights**

- The system should provide interpretable explanations for predictions using Explainable AI techniques such as SHAP.
- The explanations should highlight key features influencing the attrition risk score to improve transparency and trust in the system.

6. **Recruitment Decision Dashboard**

- The system should present results through an interactive dashboard displaying candidate rankings, skill analysis, interview evaluation results, and attrition risk scores.
- Recruiters should be able to view insights and use them to support informed hiring decisions.

5.2 Non-functional requirements

1. **Scalability**
   - The system must be scalable to handle increasing user demand without compromising performance. This includes the ability to efficiently manage resources and distribute workloads across cloud services.

2. **Performance Efficiency**
   - The system should process candidate data and generate attrition predictions within a reasonable response time to support real-time recruitment decision-making.

3. **User Interface**
   - The system must maintain a user-friendly and intuitive interface, ensuring that users can navigate the system easily and access the desired functions without confusion or difficulty.

4. **Security and Privacy**
   - The system must ensure secure handling of candidate data through encryption and access control mechanisms.

5. **Reliability and Availability**
   - The system should provide stable and consistent performance during recruitment operations, ensuring minimal downtime and reliable prediction results.

5.3 System requirements

The system requirements outline the software and hardware resources necessary for the development, deployment, and operation of the proposed conversational AI system. These requirements are critical to ensuring that the system functions properly and efficiently.

1. **Programming Language and Frameworks**
   - The system will be developed using Python, leveraging libraries such as Hugging Face Transformers for NLP tasks, and frameworks like Flask or FastAPI for creating web-based interfaces and APIs.

2. **Model Hosting**
   - The fine-tuned language models will be hosted on cloud services such as AWS or Google Cloud, utilizing containerization tools like Docker to ensure portability and scalability.

3. **Data Storage**
   - A database system, such as PostgreSQL or MongoDB, will be used to store user interactions, logs, and model-related data. The Knowledge Base will also be managed within this system, ensuring efficient retrieval and updating of information.

4. **Visualization Tools**
   - Plotting libraries such as Matplotlib or Seaborn will be employed to visualize mathematical functions, graphs, and other relevant data as part of the user interaction process.

5. **Security Measures**
   - Implementation of secure communication protocols (e.g., HTTPS), data encryption (e.g., AES), and user authentication methods (e.g., OAuth2) will be required to protect user data and ensure secure interactions.

6. **Development Environment**

- The development process will utilize IDEs like PyCharm or Visual Studio Code, along with version control systems such as Git for collaborative development and code management.

5.4 User requirements

The proposed system is designed to support different stakeholders involved in the recruitment process. Understanding their requirements is essential for ensuring that the platform meets practical industry needs.

Recruiters and HR Professionals

Recruiters and HR managers will use the system to analyze candidate profiles, evaluate job–candidate compatibility, and assess the risk of early employee resignation. The system should provide clear visual insights, candidate rankings, and interpretable AI explanations that support data-driven hiring decisions.

Hiring Managers

Hiring managers will rely on the system to review candidate evaluations and obtain additional insights regarding candidate suitability and potential retention risks. The platform should present information in a concise and understandable format to assist managerial decision-making.

6. Commercialization Plan

The proposed AI-Driven Intelligent Recruitment Decision Platform has strong commercial potential as organizations increasingly adopt artificial intelligence to improve hiring decisions and workforce management. The platform aims to provide recruitment intelligence by predicting early employee attrition risks and assisting recruiters in making data-driven hiring decisions.

6.1 Target Market and Customer Persona

The primary target market for the proposed system includes medium-to-large organizations, recruitment agencies, and human resource departments that handle high volumes of recruitment activities. These organizations often experience significant challenges related to employee turnover and inefficient hiring processes.

A typical customer persona for the system is "HR Manager or Recruitment Specialist." This individual is responsible for screening candidates, conducting interviews, and selecting suitable employees for organizational roles. The recruiter often faces challenges such as evaluating large numbers of candidates, identifying the best talent, and predicting long-term employee retention.

The proposed platform will assist these users by providing AI-based insights on candidate suitability and potential attrition risks, enabling more efficient and informed recruitment decisions.

6.2 Value Proposition

The key value proposition of the proposed platform lies in its ability to enhance recruitment decision-making through predictive analytics and explainable AI insights. Unlike traditional recruitment systems that focus only on matching skills with job requirements, this system evaluates both candidate suitability and the likelihood of long-term retention.

The platform provides several benefits for organizations, including:

- Reducing recruitment costs associated with early employee turnover
- Improving hiring efficiency through AI-based candidate ranking
- Providing interpretable insights that support transparent decision-making

By integrating predictive analytics into recruitment workflows, the platform enables organizations to move from reactive hiring strategies to proactive talent management.

6.3 Revenue Model

The proposed system can be commercialized using a Software-as-a-Service (SaaS) subscription model. Organizations can subscribe to the platform and pay a monthly or annual fee to access its features.

Possible revenue streams include:

- Subscription-based licensing for HR departments and recruitment agencies
- Enterprise packages for large organizations requiring customized integrations
- Pay-per-analysis services, where organizations pay for each candidate evaluation
- Consulting services for companies requiring advanced HR analytics or system integration

This flexible revenue model allows the platform to serve both small organizations and large enterprises.

6.4 Cost Estimation

The initial development and deployment costs of the system mainly involve cloud infrastructure, software development resources, and machine learning model training.

Key cost components include:

- Cloud computing infrastructure for hosting AI models and databases
- Data storage and processing resources
- Software development and maintenance costs
- System integration with recruitment platforms
- Security and compliance implementation

Although initial development costs may be moderate, the scalability of cloud infrastructure and SaaS deployment allows the platform to serve multiple clients while minimizing operational

expenses.

7. GANTT CHART

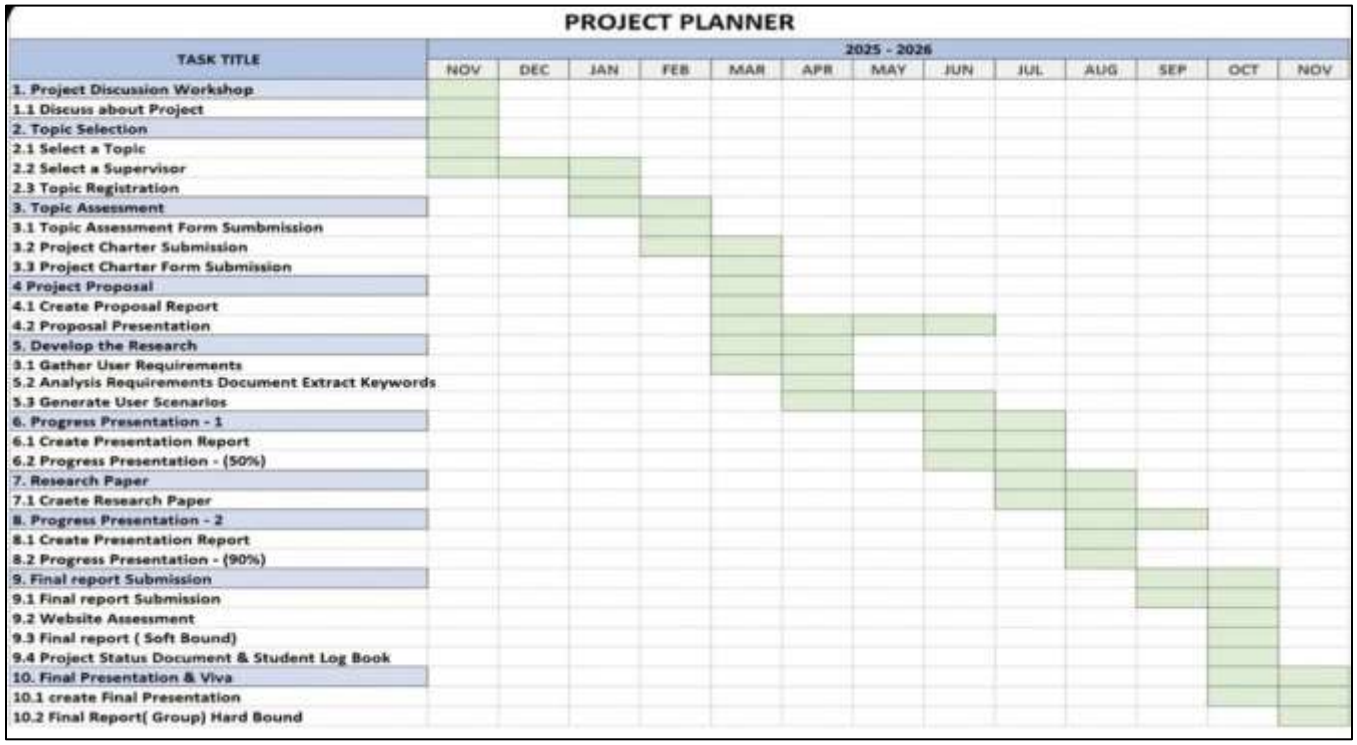


Figure 3: Gantt chart

8. Work Breakdown Structure (WBS)

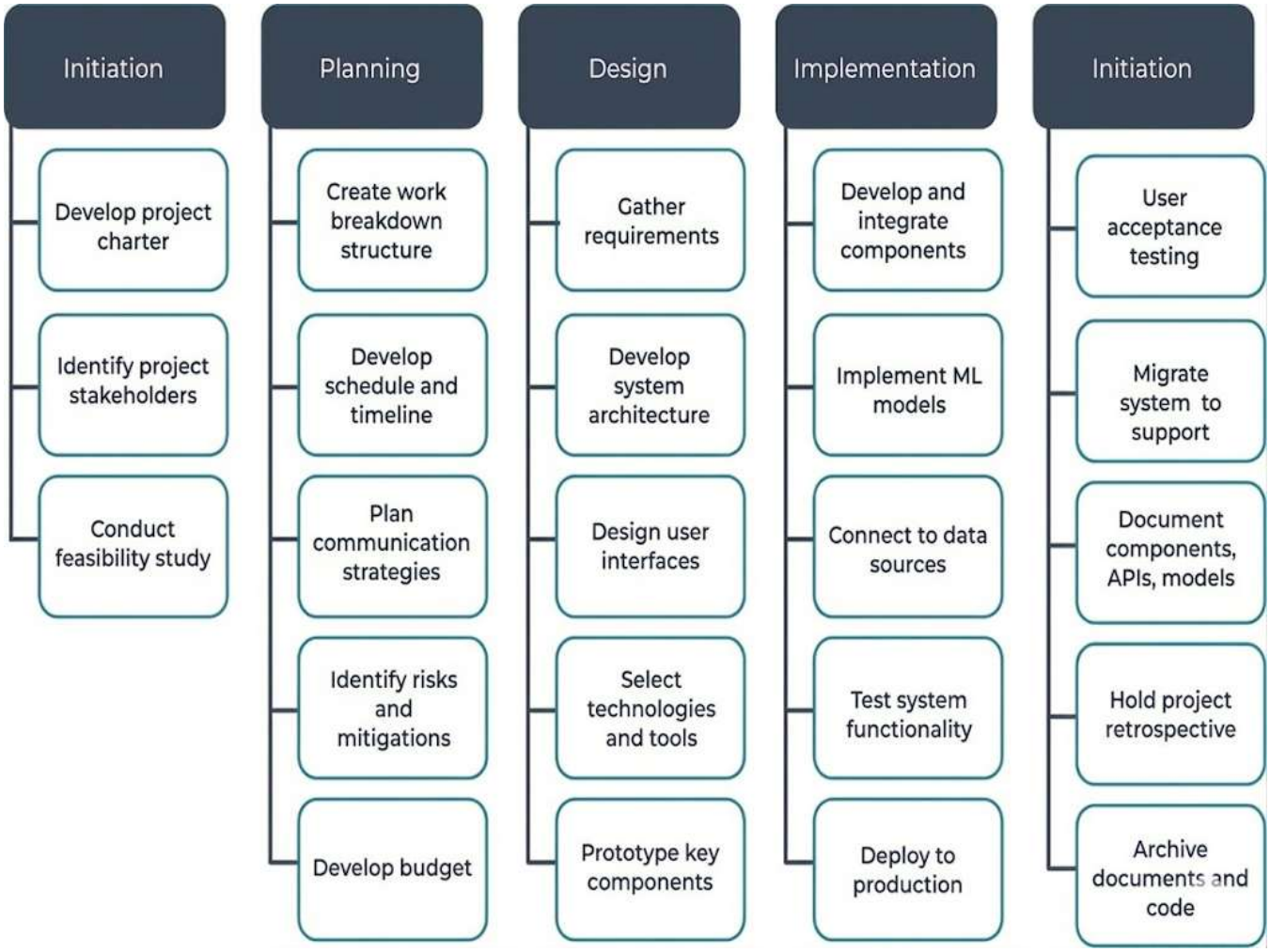


Figure 4: Work Breakdown Structure

9. ESTIMATED EXPENSES

Category	Requirement / Resource	Description	Estimated Cost (LKR)
Hardware	Development Laptop / Workstation	Used for machine learning model development, coding, and testing the recruitment AI system	250,000
Hardware	External Storage (SSD/HDD)	Storage for datasets, trained models, and project backups	20,000
Software	Development Tools (Python, VS Code, PyCharm)	Programming environment and development IDEs used to build the system	0 (Open Source)
Software	Machine Learning Libraries	Libraries such as Scikit-learn, Pandas, NumPy, and XGBoost for predictive modeling	0 (Open Source)
Software	Visualization Libraries	Tools such as Matplotlib or Plotly used for data visualization in the dashboard	0 (Open Source)
Cloud Services	Cloud Server Hosting	Cloud computing infrastructure (AWS / Google Cloud) for hosting AI models and APIs	30,000
Cloud Services	Cloud Database & Storage	Storage for candidate data, prediction outputs, and system logs	20,000
Human Effort	Research and Development Work	System design, implementation, and evaluation performed as part of academic research	0
Total Estimated Cost			**320,000 LKR**

Table 2: Estimated expenses of the proposed system

The majority of the cost is allocated to hardware resources and cloud infrastructure, which are required for training machine learning models and hosting the recruitment platform. Software tools and machine learning libraries are open-source, minimizing development costs while still providing powerful capabilities for data analysis, predictive modeling, and system deployment.

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